

Investigating the bidirectional relationships between social and emotional learning and academic outcomes among refugee students

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Abstract

This study investigates bidirectional relationships between Syrian refugee children's social and emotional learning (SEL) skills and academic performance in an education support program in Lebanon. Using data from a cluster randomized control trial with 2,757 students who completed performance-based academic assessments and reports of their SEL skills from 131 teachers, we examine bidirectional associations between academic and five teacher-observable SEL skills/behaviors: Working Memory Functioning, Inhibitory Control Functioning, Prosocial Behavior, Hyperactivity & Externalizing Behavior, and Emotional Distress & Internalizing Behavior. Controlling for child-, teacher-, household risk-, and site-level covariates, multilevel modeling revealed that Working Memory Functioning uniquely stood apart from other SEL domains: start-of-year teacher-reported Working Memory Functioning predicted end-of-year academic performance, and start-of-year academic performance predicted end-of-year teacher-reported Working Memory Functioning. No other SEL skills demonstrated such bidirectional associations. Start-of-year academic performance also negatively predicted end-of-year teacher-reported Emotional Distress. Findings highlight the distinct and central role of Working Memory Functioning in academic learning, suggesting that easier-to-collect academic performance data may serve as an indicator of SEL skills that may be difficult to assess sans strong teacher–student relationships. Further, they highlight the utility of teacher-reported SEL assessments in resource-constrained contexts, calling attention to bidirectional developmental pathways when supporting children's holistic well-being.

Keywords

Social and emotional learning, refugee well-being, holistic development, academic success, developmental pathways

Introduction

Learning experiences in schools are crucial for children who have experienced displacement and adversities, such as refugee children, by offering stable, structured environments to develop academic, social and emotional learning (SEL) competencies. These skills can foster resilience and support adaptation in post-settlement contexts (Ungar, 2012). Yet, common challenges refugee children experience in such contexts, including disrupted schooling and punitive discipline (Bahou & Zakharia, 2019; Wong & Schweitzer, 2017), often increase risk for academic, SEL and mental health difficulties. Recognizing the developmental risks faced by refugee students and the importance of both academic and SEL competencies in navigating them, there is growing interest in programs that strengthen these core capacities to support adaptation and resilience in host countries.

Though the interlinked nature of SEL skills and behaviors calls for practicing a “whole-child approach” in identifying pathways toward resilience, most research on refugee populations has

overlooked the dynamic interplay between social, emotional, cognitive, and academic learning, often emphasizing either psychosocial or academic outcomes (Kim et al., 2020, 2024; Lasater et al., 2022). Moreover, existing SEL research typically frames SEL competencies primarily as drivers of academic success (Peng & Kievit, 2020) and has been conducted largely in Minority World contexts (Global North or first world; Punch & Tisdall, 2012). While consistent with developmental cascade theory in that functioning in one domain shapes functioning in another over time (Masten & Cicchetti, 2010), this often ignores a key feature of cascades: bidirectionality (Cipriano et al., 2023; Durlak

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et al., 2011). This study addresses these two critical gaps: first, we add to a scarce body of literature on links between SEL and academic development among refugee children resettled in lower-middle-income countries, Syrian children in Lebanon, specifically—a population and context that are largely underrepresented in the field; second, we examine short-term bidirectional relationships between SEL and academic development, contributing to the broader SEL research that predominantly explores these relationships unidirectionally.

Bidirectional Relationships Between Academic and SEL Development

Academic development can serve as both a pathway to and an indicator of resilience—the ecologically complex processes that support positive developmental trajectories (including school engagement and supportive social networks), shaped by affordances in socio-physical ecologies (Ungar, 2012). As children build academic abilities, they simultaneously draw upon and strengthen SEL capacities like working memory, inhibitory control and social problem-solving—skills essential for adaptive functioning under stress (Traub & Boynton-Jarrett, 2017; Ungar, 2012). Advances or lags in one SEL domain can cascade into academic domains; for example, difficulties with attention (Blair, 2002) and regulating emotions (Ihbour et al., 2023) can undermine academic attainment. Foundational SEL and learning capacities thus develop interdependently, with strengthening SEL “muscles” catalyzing academic gains. These patterns align with developmental cascade theory, which posits that growth or disruption in one developmental domain can influence others (directly, indirectly, or bidirectionally) and reshape their trajectories (Masten & Cicchetti, 2010). Opportunities to engage in academic and SEL development can buffer against negative cascades (e.g., the well-documented link between conduct problems and academic performance) or activate positive ones (e.g., strong cognitive to academic competencies).

Recognizing the role of cognitive, social and emotional skills to academic success after educational disruption, non-governmental organizations (NGOs) have implemented education support programs to help refugee children return to school and thrive (Culbertson & Constant, 2015; Dryden-Peterson et al., 2019; Shuayb et al., 2014). These programs support refugee children in formal schools in reaching key milestones (including catching up to same-age peers and graduating) within host-country education systems (Culbertson & Constant, 2015; Shuayb et al., 2014). They were fashioned considering the multiple barriers refugee students face that threaten their holistic well-being, in addition to extensive evidence of the effectiveness of SEL interventions in Western contexts. The mechanisms by which education support programs promote these skills, however, are little known (Graziano et al., 2007; Hashash et al., 2018; McEvoy & Welker, 2000).

Understanding these mechanisms requires closer attention to the emotional and psychological challenges refugee children face in classroom settings, given that they may undermine the very goals programs seek to advance. Internalizing behaviors, common among refugee children, can disrupt learning by impairing social, emotional and cognitive development (Reed et al., 2012; Sirin & Rogers-Sirin, 2015). The emotional distress of fleeing violence and adapting to foreign contexts can cause concentration, rumination, and self-efficacy difficulties, harming working memory and further

compounding academic challenges (Ellis & Moore, 1999; Khamis, 2019). Empirical support for how internalizing and externalizing behavior shapes students’ academic success in Majority World contexts, however, is neither consistent nor extensive, with some studies finding predictive relations (Masseti et al., 2008; Owens et al., 2012) and others finding none (Masten et al., 2005). Some research in humanitarian settings has explored SEL and academic development, but rarely bidirectionally, often focusing on a select few competencies (such as working memory and externalizing behavior), with mixed results, particularly with emotional distress and prosocial behavior. Most SEL research has taken a unidirectional approach to understanding these relationships.

SEL in Learning Contexts

While nearly three decades of research conceptualizes SEL as a broad collection of social, emotional, and cognitive capacities (Collaborative for Academic, Social, and Emotional Learning, n.d.; Durlak et al., 2011), this study focuses on classroom manifestations of five teacher-observable SEL skills/behaviors that have strong theoretical and empirical links to academic learning, primarily in Western contexts: Prosocial Behavior & Social Acceptance; Hyperactivity & Externalizing Behavior; Emotional Distress & Internalizing Behavior; Inhibitory Control Functioning; and Working Memory Functioning. In classrooms, these competencies are expressed through learning-related behaviors that shape teacher perceptions and relationships with students and their academic outcomes, and are often captured through teacher reports (Kim et al., 2021; Yan et al., 2024). These skills are especially salient for refugee students, whose development is disrupted by displacement and trauma. Related stress or post-traumatic stress disorder can compromise cognitive regulation functions, including attention and working memory (Aber et al., 2024). Prosocial behavior is vital for building supportive peer and adult relationships in unfamiliar school environments and has been linked to academic attainment (Caprara et al., 2000; Gerbino et al., 2018). Heightened internalizing and externalizing behaviors may reflect adaptive responses to adversity (El-Awad et al., 2021; Lustig et al., 2004), but may also impact adult adjustment (Masten & Cicchetti, 2010). Although studies in Majority world contexts have examined both academic and SEL development, few have done so bidirectionally. Research in Minority World contexts often demonstrates mutual reinforcement between SEL and academic development, but predominantly frames SEL as a predictor and not vice versa (Cipriano et al., 2023; Durlak et al., 2011; Peng & Kievit, 2020). Given gaps in understanding critical SEL competencies that impact learning and are particularly important for refugee students, and the predominantly unidirectional focus of existing research, this study examines the reciprocal relationships between these SEL skills and academic performance. Below, we review literature linking each of these SEL skills to academic performance.

Prosocial behavior refers to voluntary acts that benefit others, such as cooperating, kindness, or sharing (Oberle et al., 2023). In Minority World contexts, it has been linked to academic success through improved self-esteem and peer- and teacher-relationships (Caprara et al., 2000; Gerbino et al., 2018). This positive relation may stem from shared behavioral demands to demonstrate competence in each respective domain (Gerbino et al., 2018; Wentzel, 1993), particularly in early schooling when the

nature of school work may be more collaborative (Hessen & Kuncel, 2022). Teachers' perceptions of students' academic performance can shape how they perceive and interact with students (Rosenthal & Jacobson, 1968; Tenenbaum & Ruck, 2007). In Majority World contexts, teacher perceptions have been linked to both prosocial behavior and academic achievement. For example, Jadoon et al. (2022) found that teacher-student relationships and teacher-reported prosocial behavior predicted academic achievement. In contrast, Shirin (2020), while acknowledging the role of teacher-student relationships, but not directly testing them, reported negative associations between prosocial behavior and academic achievement. Notably, existing studies treat prosocial behavior as a predictor of academic achievement and, to our knowledge, none have examined this bidirectionally.

Hyperactivity and Externalizing behavior—characterized by overt, disruptive behavior that violates social norms—is linked to adolescent risk-taking and poor academic performance (Hinshaw, 1992; Thompson et al., 2011). In classrooms, this may look like restlessness, difficulty listening, or yelling at peers/teachers, which can strain teacher-student relationships (Graziano et al., 2007; Hamre et al., 2008; Hashash et al., 2018; McEvoy & Welker, 2000). The consequences of externalizing behavior for school success, social competence and internalizing behavior are a widely corroborated pattern in the developmental cascades literature, with timing of assessments being critical to identify when the processes that undermine key developmental competencies impress lasting longer-term consequences (Kulakow et al., 2024; Masten & Cicchetti, 2010). For example, in childhood, inattention and hyperactivity correlate more with academic problems, while by adolescence, antisocial behavior and delinquency may become more salient. Research from the Minority World, including a meta-analysis, suggests bidirectional associations between academic achievement and externalizing behavior, with teacher reports of externalizing behavior demonstrating consistent and robust longitudinal relations with students' academic outcomes (Van der Ende et al., 2016; Yan et al., 2024).

Experiences that cause emotional distress can elevate internalizing behaviors (El-Awad et al., 2021) and negatively impact academic achievement (Rousseau et al., 1996). In classroom settings, this may manifest as social withdrawal, emotional sensitivity or somatic complaints. Evidence on the relationship between internalizing behavior and academic achievement from Minority World contexts, however, has been mixed: some report negative (Ihbour et al., 2023) and reciprocal relations (Weidman et al., 2015), others propose cascade pathways via academic difficulties (Ackerman et al., 2007; Masten et al., 2005), and others find no reciprocal links including in studies relying on teacher-reported measures (Kulakow et al., 2024) or when contrasted with parent-reported measures (Van der Ende et al., 2016). To our knowledge, no study has examined this bidirectional relationship within refugee populations.

Working memory functioning (maintaining goal-relevant information, including remembering, manipulating, and using it to guide behavior in context) and inhibitory control functioning (stopping engagement in prepotent responses) are part of a broad set of cognitive capacities under the SEL umbrella (Brown et al., 2023; Durlak et al., 2011; Gerst et al., 2017; Kim et al., 2021; Miller-Cotto & Byrnes, 2020). Research shows that children's working memory and inhibitory control skills predict later academic outcomes (Alloway & Alloway, 2010; Morrison et al.,

2010). In classroom contexts, working memory functioning is reflected in students' abilities to retain and apply information in real time, such as recalling lists or items, following multi-step instructions, and transitioning between tasks—the skills and processes that directly help students to process information and learn in classrooms that teachers can observe. In contrast, inhibitory control functioning may manifest as listening or waiting for one's turn (Alloway & Alloway, 2010; Roncadin et al., 2007). In schools, working memory and inhibitory control functioning reciprocally promote the independent learning behavior essential to academic success (Blair, 2002; Roncadin et al., 2007). Strong inhibitory control skills can be predictive of social adaptation, protective from externalizing behavior, and influence academic engagement (Allan et al., 2014; Rhoades et al., 2009). Among cognitive skills, working memory has been uniquely linked to academic attainment, with early ratings strongly linked to later academic success (Alloway & Alloway, 2010; Blair, 2002; Wolf & McCoy, 2019).

Across Minority and Majority World contexts, studies highlight a bidirectional relationship between academic skills and working memory and inhibitory control functioning. In the United States, reading and math performance predict later working memory skills, which in turn predict future academic skills (Miller-Cotto & Byrnes, 2020; Peng et al., 2018). Sankalaite et al. (2023), for example, found that teacher-reported working memory and arithmetic were significantly and bidirectionally predictive of each other. Their findings suggest that student-teacher relationship quality is linked to students' cognitive functioning and classroom behavior, while students' cognitive functioning, in turn, shapes their relationships with teachers. Similar findings emerge from Majority World contexts: in Ghana, performance-based working memory and inhibitory control and literacy/numeracy predict each other (Wolf & McCoy, 2019); in Niger, emerging work suggests that literacy predicts later direct assessments of working memory, inhibitory control, and behavioral regulation, which then support future literacy. Relatedly, Syrian refugee children in Lebanon overage for grade (a proxy for disrupted schooling) show poorer working memory and inhibitory control skills (Kim et al., 2020). Aber et al. (2024) also found that students whose teachers rated them highly on SEL skills, including working memory functioning and regulation skills, were significantly positively related to academic performance. Interestingly, students with higher teacher ratings on working memory and inhibitory control skills were more likely to perform better in academic outcomes academic achievement, than students who scored higher on performance-based cognitive tests of working memory and inhibitory control but were not rated as highly by teachers (Aber et al., 2024). Together, these findings suggest that poor academic performance and disrupted schooling may partly reflect underlying emotional and self-regulatory challenges that shape student engagement. They also indicate that teacher reports of students' SEL skills can serve as powerful, ecologically valid indicators of cognitive functioning in real-world classroom contexts, while requiring fewer resources than performance-based measures (Kim et al., 2021).

The Study

Exploring the bidirectional relationships between children's SEL and academic learning outcomes is critical to refugee children's

holistic development, given the pre-, post-, and migration risks they experience. This study examines these relationships across an academic year among Syrian refugee children attending education support programming in Lebanon, drawing on data from a larger cluster randomized controlled trial (CRCT) study evaluating the impact of the Tutoring in a Healing Classroom (HCT) program. The CRCT tested two models of HCT: HCT-only (a climate-targeted SEL approach to support students' adaptation and retention in Lebanese public schools) and HCT + SEL, which added skills-targeted training in stress and emotion regulation and executive functioning. Sixteen weeks of the HCT-only program improved students' perceptions of their school environment and behavioral regulation, but showed no significant changes in other academic or SEL outcomes, including executive functioning, inhibitory control and internalizing symptoms (Tubbs Dolan et al., 2022). In contrast, HCT + SEL students demonstrated gains in literacy, numeracy and school perceptions, alongside higher school-related stress. Sixteen weeks of the HCT + SEL program did not have any unique impacts on SEL outcomes, beyond the HCT (Brown et al., 2023). For intervention, design and evaluation details, see Tubbs Dolan et al. (2022) and Brown et al. (2023). In this study, we use data from both treatment conditions of the larger CRCT that received HCT-only and HCT + SEL to address two research questions:

- (1) Do teachers' reports of refugee students' SEL skills and behaviors (Working Memory Functioning, Inhibitory Control Functioning, Prosocial Behavior and Social Acceptance, Hyperactivity and Externalizing Behavior, and Emotional Distress and Internalizing Behavior) at the start of the school year uniquely predict academic outcomes at the end of the year?
- (2) Does academic performance at the start of the year predict teacher reports of SEL skills/behaviors at the end of the year?

Methods

Data and Sample

Data for this study were derived from a large-scale CRCT evaluating non-formal education support programs and SEL activities for Syrian refugee children enrolled in public schools in Lebanon's Bekaa and Akkar regions in the 2016–2017 school year. The sample included 2,757 Syrian refugee children (aged 5–16; $M=8.98$, $SD=2.32$; 49.66% female) enrolled in the HCT education support program, and 131 teachers (aged 18–48; $M=27.45$; $SD=6.56$; 91.19% female) hired locally and trained by the NGO to deliver the program (Brown et al., 2023; Tubbs Dolan et al., 2022). Teacher-report data could not be collected from control sites where students were not offered and did not attend HCT classes. Therefore, data from 21 control sites were not included in the present analyses. Of the 66 treatment sites, 33 sites were assigned to the HCT-only condition and 33 sites to the HCT + SEL condition. Six HCT + SEL sites were dropped after the first half year of the program due to budgetary limitations, removing their end-of-year teacher data, and therefore dropped from this study. An additional 10 teachers and their students were excluded due to missing teacher self-report information, as teacher characteristics were key covariates for controlling bias in student assessments. On average, there were 17 students per classroom across sites and teachers

completed structured SEL rating instruments consisting of 29 Likert-type-scale items. Teachers were given clear instructions and support by program and NGO staff. There was a 17 week interval between data collection time points, allowing teachers sufficient time to become familiar with students and reducing reporting biases such as fatigue or repetitive responses. This study was approved by the New York University Institutional Review Board (NYU FY2016-1174).

Measures

SEL Skills and Behaviors. Teachers rated children's classroom SEL functioning using items from the Strength and Difficulties Questionnaire (SDQ; Goodman, 1997) and Classroom Executive Function Survey (CEFS; Jones et al., 2015). The SDQ includes five subscales (Hyperactivity, Emotional Symptoms, Conduct Problems, Peer Problems, and Prosocial Behavior). The CEFS assesses a range of executive function skills, and we used the Working Memory and Inhibitory Control Functioning subscales. The 8 CEFS items were rated on a 5-point scale ("Never," "Rarely," "Occasionally," "Frequently," "All the time") and the 18 SDQ items on a 3-point scale ("Certainly true," "Somewhat true," "Not true"). The 29-item scale was designed based on a split-sample exploratory and confirmatory factor analysis conducted by Gjicali et al. (2020). The best-fitting model suggested five dimensions: (1) Working Memory Functioning ("remembers lists of items in the correct order"; 4 CEFS items; $\alpha=.88-.91$); (2) Inhibitory Control Functioning ("uses self-control techniques"; 4 CEFS items; $\alpha=.85-.90$); (3) Prosocial behavior & Social acceptance ("shares readily with other children"; 9 SDQ items; $\alpha=.85-.86$); (4) Hyperactivity and Externalizing behaviors ("constantly fidgeting or squirming"; 4 SDQ items; $\alpha=.77-.76$); and (5) Emotional distress and Internalizing behaviors ("many worries, often seems worried"; 6 SDQ items; $\alpha=.66-.67$). Final model fit was good against empirical criteria. Treatment group, gender group, and longitudinal invariance testing showed between acceptable to good model fit against empirical criteria. Because SDQ and CEFS items were rated on different response scales, subscale summed scores were standardized into z-scores to allow for comparison in our models (see Supplemental Appendix A for all SEL items).

Academic Performance. The Annual Status of Education Report (ASER) is a national learning assessment (Banerji et al., 2013) capturing students' literacy, numeracy and second language skills that has been used and adapted in low- and middle-income countries (e.g., Tanzania, Mali, Kenya, and India). For this study in Lebanon, an international NGO adapted the literacy assessment to Arabic and the Second Language to English/French. The ASER literacy assessment entails five levels: cannot read letters (0), letters (1), words (2), a short paragraph (3) and a longer paragraph (4). The math assessment also has five levels: can't recognize numbers (0), 1-digit number recognition (1), 2-digit number recognition (2), 2-digit subtraction with borrowing (3), and 3-digit by 1-digit division (4). In both, students begin at Level 3 and are promoted, demoted, or remain at the same level, yielding scores of 0–4. Due to our interest in overall academic performance, literacy, numeracy, and second language scores were averaged into a composite score with row means. Cronbach's alpha was high for academic performance, with $\alpha=.90$ at the start of the school year and $\alpha=.86$ at the end (see Supplemental

Table 1. Descriptives and Missingness (Unimputed and Imputed Data).

Key outcomes/predictors	n	Unimputed			Imputed
		M/% (SD/SE)	Range	% missing	M/% (SD/SE)
Academic performance: T1 (composite score)	2,757	1.14 (1.13)	0 to 4	0.0	1.14 (1.13)
Inhibitory Control Functioning: T1 (z-score)	2,445	.02 (1.07)	−2.53 to 2.0	11.3	.02 (1.03)
Working Memory Functioning: T1 (z-score)	2,445	.03 (1.07)	−2.52 to 2.15	11.3	.02 (1.03)
Prosocial Behavior & Social Acceptance: T1 (z-score)	2,442	−.00 (1.00)	−3.63 to 3.53	11.4	−.00 (1.00)
Hyperactivity/Externalizing Behavior: T1 (z-score)	2,442	.06 (1.06)	−1.03 to 2.91	11.4	.02 (1.03)
Emotional Distress/Internalizing Behavior: T1 (z-score)	2,442	.09 (1.05)	−0.86 to 5.03	11.4	.04 (1.02)
Academic performance: T2 (composite score)	1,631	2.26 (1.17)	0 to 4	40.8	2.27 (1.09)
Inhibitory Control Functioning: T2 (z-score)	1,718	−.03 (1.20)	−2.98 to 2.00	37.7	−.00 (1.03)
Working Memory Functioning: T2 (z-score)	1,718	.02 (1.20)	−2.91 to 2.04	37.7	.02 (1.02)
Prosocial Behavior and Social Acceptance: T2 (z-score)	1,718	−.02 (.98)	−3.62 to 3.78	37.7	−.01 (.99)
Hyperactivity/Externalizing Behavior: T2 (z-score)	1,718	.18 (1.15)	−1.06 to 3.24	37.7	.01 (1.03)
Emotional Distress/Internalizing Behavior: T2 (z-score)	1,718	.17 (1.16)	−0.83 to 5.26	37.7	.00 (1.02)

Note. Ranges for the imputed data are not reported because they were nearly identical to those observed in the unimputed data; the maximum difference in range was 0.02 units; T1 = Time 1, T2 = Time 2.

Appendix A for Cronbach's alpha for academic performance and each SEL subscale at the start and end of the school year).

Covariates. A number of child, household, teacher and site-level covariates were selected for inclusion in the models based on theory and prior research suggesting correlations with the outcomes of interest (Brown et al., 2023; Kim et al., 2020; Tubbs Dolan et al., 2022). Child-level covariates include: gender, age, overage for grade (computed by subtracting the average age of students in a respective student's grade from the student's actual age), attendance, treatment group, and type of education support program support group. Household covariates include: whether or not the child had missed a school year, worked, or the family had moved in the last year, in addition to access to toilets, access to safe drinking water, ability to afford basics, household literacy and type of housing. Teacher-level covariates include: gender, age, experience, nationality, whether or not they taught more than 1 education support class, whether or not they taught a second language, and physical health. Finally, classes per site were included as a site-level covariate (see Supplemental Appendix B for correlations between key outcomes and predictors).

Data Analytic Strategy

Missing Information. Due to substantial missingness, with 11.3%–11.4% of start-of-year teacher-report SEL scores missing, 37.8% at the end, and 40.8% of ASER scores missing at the end of the program year (0% missing at the beginning of the school year), we used multiply imputed datasets. The fraction of missing information (FMI) for key outcomes with the largest missingness was 69, based on Little and Rubin (1987)'s formula: $m = (100 \times \text{FMI}) / (1 - \text{FMI})$. To sufficiently account for the uncertainty given sizable missing information in the dataset, we imputed and used 80 datasets in our study (see Table 1 for descriptives on unimputed and imputed datasets, along with percentages of missing data for key outcome variables in the unimputed dataset). Missing data were imputed using multiple imputation through chained models (MICE; Rubin, 1987; Van Buuren et al., 2006) using R (see Montes-de-Oca-Salinas et al., 2022 for details about imputation). This approach creates a model

for each variable with missing information and, conditional on other variables, predicts missing values using all available information. A random forest modeling approach was used to predict missing values, given its ability to account for a large number of potential predictors, interactions and non-linearities, as well as its built-in mechanism to prioritize and identify predictors' importance (Doove et al., 2014; Stekhoven & Bühlmann, 2012).

Model Testing. A series of analyses was conducted to examine associations between our predictors and key outcomes. Intraclass correlations (ICCs) from unconditional models ranged from 0.00 to 0.10 at the site level and from 0.08 to 0.41 at the classroom level, indicating meaningful clustering within classrooms and justifying the use of multilevel modeling for outcome analyses to account for random effects at classroom- and site-levels. To test the contribution of teacher-reported SEL skills and behaviors to academic performance, we regressed academic performance at the end of the program year on all five SEL skills/competencies measured at the beginning of the program year, controlling for the academic performance at the beginning of the program year and covariates. We then tested the reverse; we regressed each of the teacher-reported SEL skills at the end of the program year on academic performance measured at the beginning of the school year, controlling for SEL skills/competencies and covariates at the start of the school year. Finally, using mean-centered predictors and interaction terms for each of the SEL skills and baseline academic skills, we conducted post hoc analyses to ensure robustness of main findings and examine whether bidirectional associations varied by age, being overage for grade, gender or different start-of-year levels of academic skills/SEL. All descriptive and regression models were run using Stata version 16, with estimates pooled from the models with 80 imputed datasets using Stata's mi estimation commands.

Results

Descriptive Statistics

Table 1 presents the means/frequencies and standard deviations/standard errors for all key outcomes and predictors (corresponding statistics for covariates are in Supplemental Appendix C).

Table 2. All SEL Skills at the Start of the School Year Predicting End-of-Year Academic Performance ($n = 2,757$).

	<i>b</i>	<i>SE</i>	95% CI
Inhibitory Control Functioning: T1	.03	.02	−0.02 to 0.07
Working Memory Functioning: T1	.10***	.02	0.05 to 0.14
Prosocial Behavior & Social Acceptance: T1	.00	.02	−0.03 to 0.04
Hyperactivity/Externalizing Behavior: T1	−.00	.02	−0.04 to 0.04
Emotional Distress/Internalizing Behavior: T1	−.01	.02	−0.04 to 0.03
Academic performance: T1	.57***	.04	0.50 to 0.64
_cons	.36	.25	−0.13 to 0.86
Random effects parameters	Estimate	<i>SE</i>	95% CI
Location	.109	.06	0.04 to 0.30
Class	.159	.03	0.11 to 0.24
Residual	.705	.01	0.68 to 0.73

Note. Model controlled for covariates; see Supplemental Appendix D for all estimates; T1: Time 1; T2: Time 2; CI: confidence interval.

*** $p < .001$, ** $p < .01$, * $p < .05$.

RQ1: SEL Competencies Predicting Later Academic Performance. We first modeled all five teacher-reported SEL skills/behaviors as predictors of end-of-year academic performance. Only Working Memory Functioning was predictive of academic performance ($b = .10$, $SE = .02$, $p < .01$), above and beyond the effects of the other SEL competencies and academic performance at the beginning of the program year. The full model explained 29.6% of child-level variance, compared to the null model (see Table 2 for the model with key outcomes and predictors, and Supplemental Appendix D for the full model with covariates, along with variance explained at the classroom and site levels).

RQ2: Academic Performance at the Start of the School Year Predicting Later SEL Skills/Behavior. Next, we ran five models with academic performance at the start of the school year predicting each of the teacher reports of students' SEL skills/behavior at year's end. Increases in academic performance were positively associated with Working Memory Functioning at the end of the school year ($b = .24$, $SE = .04$, $p < .001$) and negatively associated with end-of-year Emotional Distress and Internalizing Behavior ($b = -.11$, $SE = .04$, $p < .01$). Compared to the null model, the final Working Memory Functioning model accounted for 10.6% of the variance, and the emotional distress/internalizing behavior model accounted for 2.9% of the variance at the child level. Inhibitory Control Functioning, Prosocial Behavior, and Hyperactivity and Externalizing Behavior were not significantly associated with end-of-year academic performance (see Table 3 for the model with key outcomes and predictors, and Supplemental Appendix E for the full model with covariates, along with variance explained at classroom and site levels). Post hoc moderations by age, overage for grade, gender (Supplemental Appendices F–I), initial levels of academic skills and SEL were consistent with primary models with one exception: the association between baseline start-of-year academic skills and end-of-year teacher-reported Hyperactivity and Externalizing behavior varied by initial Hyperactivity and Externalizing behavior levels ($b = .05$, $SE = .02$, $p < .03$; Supplemental Appendix J).

Discussion

Teacher reports of Syrian refugee students' Working Memory Functioning were uniquely predictive of and by academic performance across a school year. In addition, low academic performance was predictive of later Emotional Distress and Internalizing Behavior. These findings highlight interconnected developmental pathways between cognitive, academic, and emotional outcomes in a humanitarian context. While links between SEL and academic development are well-documented in Minority World contexts (often in unidirectional models; Peng & Kievit, 2020), the strength and directionality of these relationships remain underexplored in humanitarian contexts (Kim et al., 2020, 2024; Lasater et al., 2022). Our findings align with developmental cascade theory and suggest that teacher reports themselves may constitute an important developmental pathway to examine in better understanding child well-being in humanitarian contexts. We also add to emerging evidence (Wolf & McCoy, 2019), underscoring how these processes are mutually influential even under conditions of displacement/adversity and offer insight into the utility of teacher-reported student SEL for decision-making in crisis-affected educational settings, which is valuable given calls for more usable, field-feasible assessment tools (Brann et al., 2022). Below, we situate each of our findings within the broader literature and discuss implications for future research and program decision-making.

Working Memory Functioning and Academic Performance: Evidence of Reciprocal Development

Accounting for demographic and ecological factors and other SEL skills/behaviors, teacher-reported Working Memory Functioning and academic performance were reciprocally predictive over the school year. This aligns with developmental cascades theory, which emphasizes the robustness of cognitive skills in shaping developmental outcomes (Masten & Cicchetti, 2010); a pattern supported in both Minority (Alloway & Alloway, 2010; Blair, 2002; Gerst et al., 2017) and Majority World contexts

Table 3. Academic Performance at the Start of the School Year Predicting End-of-Year SEL ($n = 2,757$).

	Inhibitory control functioning			Working memory functioning			Prosocial behavior and social acceptance			Hyperactivity/Externalizing behavior			Emotional distress/internalizing behavior		
	b (SE)	95% CI		b (SE)	95% CI		b (SE)	95% CI		b (SE)	95% CI		b (SE)	95% CI	
Academic performance T1	.08 (.04)	–0.00 to 0.15		.24 (.04)***	0.16 to 0.31		.05 (.05)	–0.03 to 0.15		–.03 (.04)	–0.11 to 0.04		–.11 (.04)***	–0.19 to (–0.03)	
SEL score T1	.15 (.02)***	0.12 to 0.20		.17 (.02)***	0.14 to 0.22		.03 (.02)	–0.01 to 0.08		.19 (.02)***	0.15 to 0.23		.05 (.02)	0.01 to 0.09	
Random effects parameters	Estimate (SE)	95% CI		Estimate (SE)	95% CI		Estimate (SE)	95% CI		Estimate (SE)	95% CI		Estimate (SE)	95% CI	
Location	.28 (.04)	0.20 to 0.38		.25 (.05)	0.18 to 0.36		.02 (1.83)	1.05E–69 to 5.10E+65		.11 (.16)	0.01 to 1.89		.39 (.05)	0.30 to 0.50	
Class	.21 (.04)	0.14 to 0.30		.25 (.04)	0.19 to 0.33		.26 (.04)	0.20 to 0.35		.27 (.04)	0.20 to 0.37		.25 (.04)	0.18 to 0.33	
Residual	.90 (.01)	0.87 to 0.93		.85 (.01)	0.82 to 0.87		.94 (.02)	0.90 to 0.97		.91 (.01)	0.88 to 0.94		.88 (.01)	0.86 to 0.91	

Note. Model controlled for covariates; see Supplemental Appendix E for all estimates; T1: Time 1; CI: confidence interval.

*** $p < .001$, ** $p < .01$, * $p < .05$.

(Wolf and McCoy, 2019). Though we did not test temporal ordering, the reciprocal associations observed suggest a “snowball effect” that creates a reinforcing cycle between working memory functioning and academic development. Strong working memory functioning may support instruction processing, task management, and content retention, which promote academic achievement. In turn, academic success may strengthen cognitive capacities. Wolf and McCoy (2019) reported similar reciprocal associations between executive function and literacy and numeracy skills among Ghanaian children, and, in Niger, emerging work spotlights literacy as predictive of later executive function and behavioral regulation skills. While migration- and displacement-related stressors can hinder refugee students’ working memory functioning and academic development, findings from this and emerging studies suggest that working memory functioning remains malleable and that some of the SEL and academic risks they face can be mitigated through access to academic and SEL development opportunities, notably those that focus on working memory functioning. Our results also highlight the utility of teacher reports of working memory functioning as valuable tools that can reliably predict behavioral manifestations of it that significantly impact academic performance.

Academic Performance and Emotional Distress

Academic performance at the start of the school year was negatively associated with teacher-reported Emotional Distress and Internalizing Behavior at the end of the year. Prior research (predominantly from Minority World contexts) has largely examined internalizing behavior as a predictor of academic outcomes, yielding mixed findings (Kim et al., 2020; Masten et al., 2005). Consistent with Kim et al. (2020), we found no significant associations between students’ start-of-year Emotional Distress and Internalizing Behavior and end-of-year academic performance. However, reversing the directionality of this relationship and administering teacher reports of students’ SEL skills to the same teachers at two time points revealed that lower academic performance predicted higher Emotional Distress and Internalizing Behavior. One explanation is that teachers more accurately recognize internalizing symptoms as relationships deepen, consistent with evidence that teachers often underrate these behaviors relative to caregivers (Hinshaw, 1992; Stanger & Lewis, 1993). This may explain why start-of-year teacher reports did not predict later academic outcomes in this study and in others using cross-sectional or early-year teacher reports. For example, Van der Ende et al. (2016) found no significant bidirectional associations between teacher- or parent-reported internalizing behaviors and academic performance, noting that teacher-reported behaviors were less stable over time, partly due to structural factors like teacher turnover across school years. In contrast, our study benefits from the same teachers reporting on students’ behavior at both time points, improving their ability to reliably report on these behaviors over time. These findings suggest that underachievement in this context can serve as an early, observable marker of emotional distress that can be identified before teachers formally recognize such symptoms.

An alternative explanation is that the relationship is causal: underperformance in school may indirectly lead to emotional distress by undermining students’ self-esteem, sense of belonging, and connection with peers, teachers, or the broader school

community. For example, Ibour et al. (2023) found a cascading effect in which reading challenges among Moroccan secondary students foster feelings of inadequacy and public judgment, contributing to internalizing symptoms and diminished self-esteem. This may be especially salient for refugee students who often experience marginalization. Such dynamics could elevate internalizing behavior and further hinder both academic and SEL development.

Finally, developmental cascades are time-sensitive, making it critical to consider when and how competencies emerge, accelerate and exert lasting effects (Masten & Cicchetti, 2010). The null associations observed between academic performance and other SEL competencies may reflect developmental variation in our sample, which ranged from early-middle childhood to early adolescence. To explore this, we conducted post hoc moderation analyses testing whether age, being overage for grade, or being female moderated SEL to academic associations and vice versa (Supplemental Appendices F–J). While some interactions were marginally significant, none met the conventional threshold of $p < .05$. These exploratory results underscore the need for future research to attend more closely to developmental sensitivity when examining SEL-academic linkages. For instance, externalizing behavior is more linked to academic difficulties in childhood, whereas antisocial behavior becomes more salient in adolescence (Masten & Cicchetti, 2010). Inhibitory control functioning also develops rapidly in early childhood and later integrates with other executive functioning skills such as working memory functioning, making it harder to isolate in older students (Best et al., 2011). Further post hoc analyses also suggested heterogeneity in the co-development of academic performance and Hyperactivity and Externalizing behavior. Higher start-of-year academic performance was associated with greater reductions in teacher-reported Hyperactivity and Externalizing behavior among students with initially low levels of these behaviors, but with higher end-of-year ratings among students with higher initial levels (Appendix J). One hypothesis is that, for students with higher baseline Hyperactivity and Externalizing behavior, stronger academic performance may increase teacher attention, expectations, or interaction frequency, heightening the visibility of disruptive behaviors and influencing subsequent ratings. Although exploratory, this pattern warrants further investigation to examine how teacher attention, expectations, and classroom demands shape bidirectional academic-SEL associations that may be more distal to academic task demands.

While we found evidence that at least one of the SEL skills examined—Working Memory Functioning—has a cascading effect, this SEL skill is uniquely positioned because it is proximal to academic task demands, continuously exercised during instruction and shows rapid malleability over short intervals. Our post hoc analyses suggest that there are further patterns to be explored. Other SEL domains may influence academic outcomes indirectly, over longer periods, or through mediators not captured in a 1-year design. Thus, SEL and academic skills may be associated in indirect ways not fully explored in this study. Overall, the absence of significant main associations between academic performance, Hyperactivity and Externalizing Behavior, Prosocial Behavior and Social Acceptance and Inhibitory Control Functioning does not negate their importance; rather, it underscores the need for developmentally sensitive, multimodal approaches to assessment.

Limitations and Future Directions

This study has several limitations. First, findings are limited in generalizability to Syrian refugee children enrolled in education support programming delivered by predominantly host-country teachers. All participants in this study received the HCT or HCT + SEL intervention; while we controlled for treatment in our models to mitigate from potential bias arising from teachers serving as both interventionists and informants, confounders remain. Future studies should examine whether these dynamics vary when programming is delivered by teachers from refugee communities, in formal education settings or in different cultural contexts, and in comparison to populations who receive no SEL treatment. In addition, while we focused on an important but narrow set of SEL skills, many critical competencies remain unexplored. Future work should investigate a broader range of SEL skills that may be especially salient in these settings and conduct comparative analyses across diverse educational and displacement contexts to identify priority pathways for intervention and policy.

Second, SEL measures captured classroom-based behavior, which may not generalize to other settings. For example, Prosocial Behavior and Social Acceptance ratings assess actions like helping or sharing, which may be suppressed in classrooms that discourage speaking out or peer interaction, and so such behaviors outside school may go unnoticed. Reliance on teacher reports also means we may have captured behavioral expressions rather than underlying cognitive mechanisms (Kim et al., 2020). For instance, Working Memory Functioning was assessed through behaviors such as following multi-step instructions, while Inhibitory Control Functioning depended on teachers' perceptions of self-regulation and patience, which are harder to observe. These ratings may also be shaped by cultural expectations or Teacher Expectancy Effects (wherein internal or external stimuli trigger teachers' expectations of students, which are then communicated through verbally and non-verbally to students, who then adopt activities that conform to expectations; thereby transforming student outcomes; Rosenthal & Jacobson, 1968). Our main findings, alongside the pattern found in our exploratory pos-hoc analysis wherein Hyperactivity & Externalizing behavior moderated the relationship between academic performance and later hyperactivity, suggest that future research should conceptualize developmental cascades as ecologically embedded processes that unfold through interactions between children and their learning environments and examine how teacher expectations and classroom demands shape bidirectional academic-SEL associations. In classroom settings, teacher perceptions and instructional practices serve as key mechanisms linking observed SEL-related behaviors and academic outcomes. Multi-informant designs incorporating caregiver and student perspectives across home and school settings will also be critical to discerning whether the mixed or null bidirectional findings between SEL and academic outcomes reflect true developmental patterns or measurement limitations. These lines of inquiry are particularly important in crisis-affected contexts, where access to direct assessments may be limited, yet decisions about children's learning and well-being are often made based on observable behavior in educational settings. Nonetheless, when teacher reports predict academic performance, they provide a practical, scalable method for assessing observable SEL indicators in low-resource, crisis-affected contexts.

Finally, despite using 80 imputed datasets and conducting robustness checks, the high level of missing data remains a significant limitation. Multiple imputation assumes data are missing at random. If this assumption is incorrect, imputation may not fully address bias, limiting the precision and generalizability of our findings, especially for those most at risk.

Implications and Conclusion

Our findings have several implications for practice, policy and research. First, it adds to a large body of research in the Majority World emphasizing the importance of working memory functioning in academic learning and a smaller, but growing body of research in the Minority World that does the same. By situating these associations within the lived realities of Syrian refugee students in classrooms and emphasizing contextually-grounded assessments, this study contributes to a more globally-inclusive understanding of SEL-academic co-development. Second, it demonstrates that teachers in humanitarian settings can use everyday classroom behaviors, like difficulties following instructions or transitioning between tasks, as early signs of working memory challenges that may impede academic learning. In addition, poor academic performance at the start of the school year may serve as a low-barrier screening tool to identify students at risk of internalizing distress. Teacher training to recognize the quiet signs of emotional distress (including somatic complaints, appearing worried, or being bullied) in students who may not act out can help identify those who may be struggling academically. Third, because teacher perceptions shape expectations and student-teacher interactions, reciprocal associations between teacher-reported SEL behaviors and academic performance may themselves constitute an important developmental pathway. More broadly, our findings underscore that access to education is not merely beneficial but *essential* for refugee children's academic and holistic development. They call for a shift from unidirectional models toward approaches that recognize the reciprocal influences between academic and SEL development, shaped by developmental timing. While academic achievement is undoubtedly a key predictor of long-term outcomes such as school completion, higher education, and employment, it is equally important to attend to broader developmental competencies that allow for students, particularly those who have faced profound adversity, to lead healthy, fulfilling lives beyond academic and occupational domains. Academic success contributes to this trajectory, but so too does the capacity for critical thinking, emotional regulation, and the development of meaningful interpersonal relationships. These findings call for rethinking the outcomes we prioritize to ensure that our research, policies, and practices reflect a broader vision of what it means to thrive.

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Supplemental Material

Supplemental material for this article is available online.

References

- Aber, J. L., Kim, H. Y., & Wu, Z. (2024). Patterns of self-regulation and emotional well-being among Syrian refugee children in Lebanon: An exploratory person-centered approach. *Development and Psychopathology*, 36(5), 2522–2541.
- Ackerman, B. P., Izard, C. E., Kobak, R., Brown, E. D., & Smith, C. (2007). Relation between reading problems and internalizing behavior in school for preadolescent children from economically disadvantaged families. *Child Development*, 78(2), 581–596.
- Allan, N. P., Hume, L. E., Allan, D. M., Farrington, A. L., & Lonigan, C. J. (2014). Relations between inhibitory control and the development of academic skills in preschool and kindergarten: A meta-analysis. *Developmental Psychology*, 50(10), 2368.
- Alloway, T. P., & Alloway, R. G. (2010). Investigating the predictive roles of working memory and IQ in academic attainment. *Journal of Experimental Child Psychology*, 106(1), 20–29.
- Bahou, L., & Zakharia, Z. (2019). 'Maybe that's how they learned in the past, but we don't learn like this today': Youth perspectives on violent discipline in Lebanon's public schools. *International Journal of Educational Development*, 70, 102098.
- Banerji, R., Bhattacharjee, S., Wadhwa, W. (2013). The annual status of education report (ASER). *Research in Comparative and International Education*, 8(3), 387–396.
- Best, J. R., Miller, P. H., & Naglieri, J. A. (2011). Relations between executive function and academic achievement from ages 5 to 17 in a large, representative national sample. *Learning and Individual Differences*, 21(4), 327–336.
- Blair, C. (2002). School readiness: Integrating cognition and emotion in a neurobiological conceptualization of children's functioning at school entry. *American Psychologist*, 57(2), 111–127.
- Brann, K. L., Daniels, B., Chafouleas, S. M., & DiOrio, C. A. (2022). Usability of social, emotional, and behavioral assessments in schools: A systematic review from 2009 to 2019. *School Psychology Review*, 51(1), 6–24.

- Brown, L., Gjicali, K., Kim, H. Y., Tubbs Dolan, C., Frisoli, P., Bwary, M., & Aber, J. L. (2023). Evaluating program enhancement strategies for remedial tutoring: A cluster-randomized control trial with Syrian refugee students in Lebanon. *AERA Open*, 9, 23328584231209268.
- Caprara, G. V., Barbaranelli, C., Pastorelli, C., Bandura, A., & Zimbardo, P. G. (2000). Prosocial foundations of children's academic achievement. *Psychological Science*, 11(4), 302–306.
- Cipriano, C., Strambler, M. J., Naples, L. H., Ha, C., Kirk, M., Wood, M., Sehgal, K., Zieher, A. K., Eveleigh, A., McCarthy, M., Funaro, M., Ponnock, A., Chow, J. C., & Durlak, J. (2023). The state of evidence for social and emotional learning: A contemporary meta-analysis of universal school-based SEL interventions. *Child Development*, 94(5), 1181–1204. <https://doi.org/10.1111/cdev.13968>
- Collaborative for Academic, Social, and Emotional Learning. (n.d.). *Our history*. CASEL. <https://casel.org/about-us/our-history/>
- Culbertson, S., & Constant, L. (2015). *Education of Syrian refugee children: Managing the crisis in Turkey, Lebanon, and Jordan*. RAND Corporation.
- Doove, L. L., Van Buuren, S., & Dusseldorp, E. (2014). Recursive partitioning for missing data imputation in the presence of interaction effects. *Computational Statistics & Data Analysis*, 72, 92–104.
- Dryden-Peterson, S., Adelman, E., Bellino, M. J., & Chopra, V. (2019). The purposes of refugee education: Policy and practice of including refugees in national education systems. *Sociology of Education*, 92(4), 346–366. <https://doi.org/10.1177/0038040719863054>
- Durlak, J. A., Weissberg, R. P., Dymnicki, A. B., Taylor, R. D., & Schellinger, K. B. (2011). The impact of enhancing students' social and emotional learning: A meta-analysis of school-based universal interventions. *Child Development*, 82(1), 405–432.
- El-Awad, U., Reinelt, T., Braig, J., Nilles, H., Kerkhoff, D., Schmees, P., Rueth, J.-E., Fathi, A., Vasileva, M., Petermann, F., Eschenbeck, H., & Lohaus, A. (2021). Polytraumatization in young male refugees from the Middle East and its association with internalizing and externalizing symptoms. *Child and Adolescent Psychiatry and Mental Health*, 15(1), 75.
- Ellis, H. C., & Moore, B. A. (1999). Mood and memory. In T. Dalgleish & M. J. Power (Eds.), *Handbook of cognition and emotion* (pp. 193–210). John Wiley.
- Gerbino, M., Zuffianò, A., Eisenberg, N., Castellani, V., Luengo Kanacri, B. P., Pastorelli, C., & Caprara, G. V. (2018). Adolescents' prosocial behavior predicts good grades beyond intelligence and personality traits. *Journal of Personality*, 86(2), 247–260.
- Gerst, E. H., Cirino, P. T., Fletcher, J. M., & Yoshida, H. (2017). Cognitive and behavioral rating measures of executive function as predictors of academic outcomes in children. *Child Neuropsychology*, 23(4), 381–407.
- Gjicali, K., Wu, Z., & Kim, H. Y. (2020). *Education in emergencies: Evidence for Action (3EA) Lebanon 2016-17: Psychometric report of measures for academic, cognitive, social and emotional skills*. [Technical report]. figshare. <https://doi.org/10.6084/m9.figshare.24107898.v1>
- Goodman, R. (1997). The strengths and difficulties questionnaire: A research note. *Journal of Child Psychology and Psychiatry*, 38(5), 581–586.
- Graziano, P. A., Reavis, R. D., Keane, S. P., & Calkins, S. D. (2007). The role of emotion regulation in children's early academic success. *Journal of School Psychology*, 45(1), 3–19.
- Hamre, B. K., Pianta, R. C., Downer, J. T., & Mashburn, A. J. (2008). Teachers' perceptions of conflict with young students: Looking beyond problem behaviors. *Social Development*, 17(1), 115–136.
- Hashash, M., Abouchedid, K., & Abourjeily, S. (2018). Student-teacher interaction in public schools in Lebanon: A symbolic interactionist perspective in grade 6 classes. *SAGE Open*, 8(2), 2158244018783039.
- Hessen, P. R., & Kuncel, N. R. (2022). Beyond grades: A meta-analysis of personality predictors of academic behavior in middle school and high school. *Personality and Individual Differences*, 199, 111809.
- Hinshaw, S. P. (1992). Externalizing behavior problems and academic underachievement in childhood and adolescence: Causal relationships and underlying mechanisms. *Psychological Bulletin*, 111(1), 127–155.
- Ihbour, S., Essaidi, O., Laaroussi, M., Najimi, M., & Chigr, F. (2023). Links between reading acquisition level, emotional difficulties and academic performance in school-aged children. *The Journal of Mental Health Training, Education and Practice*, 18(2), 135–145.
- Jadoon, A. I., Khan, F., Syeda Tehmina Naz Bukhari, N. T. S., Gilani, S. Z., Ishfaq, U., & Ullah, M. (2022). Effect of teacher-student relationship on pro-social behavior and academic achievement of secondary school students. *Indian Journal of Economics and Business*, 21(1), 331–337.
- Jones, S. M., Bailey, R., Barnes, S. (2015). *Classroom Executive Function Survey (CEFS): A measure of executive function and self-regulation skills in everyday classroom behavior*. Harvard University.
- Khamis, V. (2019). Posttraumatic stress disorder and emotion dysregulation among Syrian refugee children and adolescents resettled in Lebanon and Jordan. *Child Abuse & Neglect*, 89, 29–39.
- Kim, H. Y., Brown, L., Dolan, C. T., Sheridan, M., & Aber, J. L. (2020). Post-migration risks, developmental processes, and learning among Syrian refugee children in Lebanon. *Journal of Applied Developmental Psychology*, 69, 101142.
- Kim, H. Y., Deitz, R., & Aber, J. L. (2024). SEL in countries affected by armed conflicts and protracted crises. In J. A. Durlak, C. E. Domitrovich, & J. L. Mahoney (Eds.), *Handbook of social and emotional learning* (2nd ed., pp. 542–556). Guilford Press. [www.doi.org/10.6084/m9.figshare.24107691](https://doi.org/10.6084/m9.figshare.24107691)
- Kim, H. Y., Gjicali, K., Wu, Z., & Tubbs Dolan, C. (2021). Teachers' observations of learners' social and emotional learning: Psychometric evidence for program evaluation in education in emergencies. *Journal on Education in Emergencies*, 7(2), 57–103. <https://doi.org/10.33682/3nr1-3ksq>
- Kulakow, S., Mahlau, K., & Kocaj, A. (2024). The longitudinal relationship between internalizing and externalizing behavioral problems with academic achievement in elementary school. *Learning and Instruction*, 92, 101909.
- Lasater, M. E., Flemming, J., Bourey, C., Nemiro, A., & Meyer, S. R. (2022). School-based MHPSS interventions in humanitarian contexts: A realist review. *BMJ Open*, 12, e054856. <https://doi.org/10.1136/bmjopen-2021-054856>
- Lustig, S. L., Kia-Keating, M., Knight, W. G., Geltman, P., Ellis, H., Kinzie, J. D., Keane, T., & Saxe, G. N. (2004). Review of child and adolescent refugee mental health. *Journal of the American Academy of Child & Adolescent Psychiatry*, 43(1), 24–36.
- Massetti, G. M., Lahey, B. B., Pelham, W. E., Loney, J., Ehrhardt, A., Lee, S. S., & Kipp, H. (2008). Academic achievement over 8 years among children who met modified criteria for attention-deficit/hyperactivity disorder at 4–6 years of age. *Journal of Abnormal Child Psychology*, 36, 399–410.

- Masten, A. S., & Cicchetti, D. (2010). Developmental cascades. *Development and Psychopathology*, 22(3), 491–495.
- Masten, A. S., Roisman, G. I., Long, J. D., Burt, K. B., Obradović, J., Riley, J. R., Boelke Stennes, K., & Tellegen, A. (2005). Developmental cascades: Linking academic achievement and externalizing and internalizing symptoms over 20 years. *Developmental Psychology*, 41(5), 733–746.
- McEvoy, A., & Welker, R. (2000). Antisocial behavior, academic failure, and school climate: A critical review. *Journal of Emotional and Behavioral Disorders*, 8(3), 130–140.
- Miller-Cotto, D., & Byrnes, J. P. (2020). What's the best way to characterize the relationship between working memory and achievement?: An initial examination of competing theories. *Journal of Educational Psychology*, 112(5), 1074–1084.
- Montes-de-Oca-Salinas, M. M., Hill, J., Aber, L., Dolan, C. T., & Gjicali, K. (2022). The impact of attending a remedial support program on Syrian children's reading skills: Using BART for causal inference. arXiv preprint arXiv:2208.13906.
- Morrison, F. J., Ponitz, C. C., & McClelland, M. M. (2010). Self-regulation and academic achievement in the transition to school. In S. D. Calkins & M. A. Bell (Eds.), *Child development at the intersection of emotion and cognition* (pp. 203–224). American Psychological Association.
- Oberle, E., Ji, X. R., & Molyneux, T. M. (2023). Pathways from prosocial behaviour to emotional health and academic achievement in early adolescence. *The Journal of Early Adolescence*, 43(5), 632–653.
- Owens, M., Stevenson, J., Hadwin, J. A., & Norgate, R. (2012). Anxiety and depression in academic performance: An exploration of the mediating factors of worry and working memory. *School Psychology International*, 33(4), 433–449.
- Peng, P., Barnes, M., Wang, C., Wang, W., Li, S., Swanson, H. L., Dardick, W., & Tao, S. (2018). A meta-analysis on the relation between reading and working memory. *Psychological Bulletin*, 144(1), 48–76.
- Peng, P., & Kievit, R. A. (2020). The development of academic achievement and cognitive abilities: A bidirectional perspective. *Child Development Perspectives*, 14(1), 15–20. <https://doi.org/10.1111/cdep.12352>
- Punch, S., & Tisdall, E. K. M. (2012). Exploring children and young people's relationships across majority and minority worlds. *Children's Geographies*, 10(3), 241–248.
- Reed, R. V., Fazel, M., Jones, L., Panter-Brick, C., & Stein, A. (2012). Mental health of displaced and refugee children resettled in low-income and middle-income countries: Risk and protective factors. *The Lancet*, 379(9812), 250–265.
- Rhoades, B. L., Greenberg, M. T., & Domitrovich, C. E. (2009). The contribution of inhibitory control to preschoolers' social-emotional competence. *Journal of Applied Developmental Psychology*, 30(3), 310–320.
- Roncadin, C., Pascual-Leone, J., Rich, J. B., & Dennis, M. (2007). Developmental relations between working memory and inhibitory control. *Journal of the International Neuropsychological Society*, 13(1), 59–67.
- Rosenthal, R., & Jacobson, L. (1968). *Pygmalion in the classroom: Teacher expectations and student intellectual development*. Holt.
- Rousseau, C., Drapeau, A., & Corin, E. (1996). School performance and emotional problems in refugee children. *American Journal of Orthopsychiatry*, 66(2), 239–251.
- Rubin, D. B. (1987). *Multiple imputation for survey nonresponse*. Wiley.
- Sankalaite, S., Huizinga, M., Warreyn, P., Dewandeleer, J., & Baeyens, D. (2023). The association between working memory, teacher-student relationship, and academic performance in primary school children. *Frontiers in Psychology*, 14, 1240741.
- Shirin, A. (2020). Determining the relationship between academic achievement and prosocial behavior of secondary school students in Dhaka City. *International Journal of Research and Reviews in Education*, 6(1), 6–15.
- Shuayb, M., Makkouk, N., & Tuttunji, S. (2014). *Widening access to quality education for Syrian refugees: The role of private and NGO sectors in Lebanon*. Centre for Lebanese Studies.
- Sirin, S. R., & Rogers-Sirin, L. (2015). *The educational and mental health needs of Syrian refugee children* (p. 13). Migration Policy Institute.
- Stanger, C., & Lewis, M. (1993). Agreement among parents, teachers, and children on internalizing and externalizing behavior problems. *Journal of Clinical Child Psychology*, 22(1), 107–116.
- Stekhoven, D. J., & Bühlmann, P. (2012). MissForest—non-parametric missing value imputation for mixed-type data. *Bioinformatics*, 28(1), 112–118.
- Tenenbaum, H. R., & Ruck, M. D. (2007). Are teachers' expectations different for racial minority than for European American students? A meta-analysis. *Journal of Educational Psychology*, 99(2), 253–273.
- Thompson, R., Tabone, J. K., Litrownik, A. J., Briggs, E. C., Hussey, J. M., English, D. J., & Dubowitz, H. (2011). Early adolescent risk behavior outcomes of childhood externalizing behavioral trajectories. *The Journal of Early Adolescence*, 31(2), 234–257.
- Traub, F., & Boynton-Jarrett, R. (2017). Modifiable resilience factors to childhood adversity for clinical pediatric practice. *Pediatrics*, 139(5), e20162569.
- Tubbs Dolan, C., Kim, H. Y., Brown, L., Gjicali, K., Borsani, S., Houchaimi, S. E., & Aber, J. L. (2022). Supporting Syrian refugee children's academic and social-emotional learning in national education systems: A cluster randomized controlled trial of non-formal remedial support and mindfulness programs in Lebanon. *American Educational Research Journal*, 59(3), 419–460.
- Ungar, M. (2012). Social ecologies and their contribution to resilience. In M. Ungar (Ed.), *The social ecology of resilience: A handbook of theory and practice* (pp. 13–31). Springer. https://doi.org/10.1007/978-1-4614-0586-3_2
- Van Buuren, S., Brand, J. P., Groothuis-Oudshoorn, C. G., & Rubin, D. B. (2006). Fully conditional specification in multivariate imputation. *Journal of Statistical Computation and Simulation*, 76(12), 1049–1064.
- Van der Ende, J., Verhulst, F. C., & Tiemeier, H. (2016). The bidirectional pathways between internalizing and externalizing problems and academic performance from 6 to 18 years. *Development and Psychopathology*, 28(3), 855–867.
- Weidman, A. C., Augustine, A. A., Murayama, K., & Elliot, A. J. (2015). Internalizing symptomatology and academic achievement: Bi-directional prospective relations in adolescence. *Journal of Research in Personality*, 58, 106–114.
- Wentzel, K. R. (1993). Does being good make the grade? Social behavior and academic competence in middle school. *Journal of Educational Psychology*, 85(2), 357–364.
- Wolf, S., & McCoy, D. C. (2019). The role of executive function and social-emotional skills in the development of literacy and numeracy during preschool: A cross-lagged longitudinal study. *Developmental Science*, 22(4), e12800.
- Wong, C. W., & Schweitzer, R. D. (2017). Individual, premigration and postsettlement factors, and academic achievement in adolescents from refugee backgrounds: A systematic review and model. *Transcultural Psychiatry*, 54(5-6), 756–782.
- Yan, S., Liu, Z., Peng, P., & Yan, N. (2024). The reciprocal relations between externalizing behaviors and academic performance among school-aged children: A meta-analysis of longitudinal studies. *Educational Psychology Review*, 36(4), 120.